



Glyphosate, steam and cutting for non-native plant control in Alberta fescue grassland restoration

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Keywords

Diversity; Ecological Restoration; Forb; Grass; Herbicide; Invasive; Native Species; Recovery

Nomenclature

Kuijt (1982), Moss (1983)

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Abstract

Question: What are the effects of cutting, glyphosate application and steam application on abundance and diversity of non-native grasses and forbs and non-target native grasses and forbs in restoration of a complex disturbed fescue grassland?

Location: Waterton Lakes National Park, Alberta, Canada.

Methods: Cutting, glyphosate application and steam application treatments were implemented at three disturbed sites in an incomplete block design with a control. Plant communities were evaluated for four growing seasons, one before and three after management treatment implementation.

Results: Glyphosate reduced non-native grass cover for three growing seasons following application and non-native forbs for one growing season. Glyphosate led to significant increases in non-native forb cover, more than double pre-existing values 2 and 3 yr after application. Native species abundance and diversity were more negatively impacted by glyphosate on sites with higher abundance and diversity prior to management treatments. Low frequency cutting over 2 yr did not consistently control non-native species, steam reduced non-native grass cover at the most heavily invaded site.

Conclusions: Site-specific conditions must be considered to develop effective control methods for non-native species. No treatment effectively re-established native grassland communities. Glyphosate application reduced non-native grasses, but not non-native forbs. When native forbs were abundant prior to management, glyphosate reduced them. Steam may have potential and should be further investigated. This is one of only a few studies to investigate methods to manage multiple non-native species occurring with native species, rather than management of a single undesirable species, and is the first to assess steam as a management option for native grasslands.

Introduction

Native grasslands are globally diverse, highly endangered ecosystems (Gibson 2009). In central Canada, approximately 70% of native grasslands and 83% of foothills fescue grasslands have been lost due to agriculture, human disturbances, woody species encroachment, fire suppression, plains bison extirpation and invasion by non-native species (Adams et al. 2009; Government of Canada 2010). Maintenance of remaining grasslands and restoration of disturbed grasslands are the primary focus of conservation efforts. Non-native plants are a major issue in ecological restoration because of their aggressive competition with native plants, ability to proliferate in disturbed

environments, plentiful seed source from urban and agricultural areas and persistent seed banks (D'Antonio & Meyerson 2002; Pyšek et al. 2012). Non-native species targeted for eradication dominate some disturbed sites, although their distribution is most often patchy and interspersed with native species. Methods to control non-native species are often not appropriate in restoration as they can negatively affect native species to be retained or re-introduced (Powers et al. 2013; Skurski et al. 2013; Wagner & Nelson 2014). The response of native plants to management is not well known for most methods. Non-native control methods may only be detrimental to native species in the short term and dependent on timing of management application (Adams et al. 2014; Martin et al. 2014;

Kaiser-Bunbury et al. 2015; Munson et al. 2015). Simple and cost-effective methods with low effect on native species and high control of non-native species are needed.

Most non-native plant management research in grasslands has targeted a single species, such as *Agropyron cristatum* L. (crested wheat grass; Hansen & Wilson 2006) or *Bromus tectorum* L. (downy brome; Munson et al. 2015), although there is usually a diverse assemblage of non-native and native species in grasslands slated for restoration. Land managers are thus challenged with multiple non-native species widespread within sites and low budgets. Managers mostly target sites dominated by non-natives as they are more obvious and potentially easier to address. However, sites interspersed with native species may have higher conservation value (Wiens & Hobbs 2015).

Herbicides are commonly used in restoration for difficult to control species or when mechanical control is too damaging. Herbicide treatments are applied to target species or sometimes non-selective treatments are applied over a management area prior to revegetation. Glyphosate (N-phosphonomethyl glycine) is used in restoration worldwide as it is broad spectrum and inexpensive, with low toxicity to most organisms (Giesy et al. 2000). Broadcast application can increase native species abundance and richness when competition from non-native species is eliminated or reduced (Wilson & Pärtel 2003; Simmons et al. 2007; Bahm et al. 2011a,b; Orloff et al. 2015; Tucker et al. 2017). Others report herbicides can control target species, but have few positive effects on non-target native species (Dickens et al. 2016). Non-target plant community level responses to herbicide during restoration are understudied although this information is needed to evaluate the trade-offs between target and non-target species where restoration is the goal (Skurski et al. 2013; Martin et al. 2014). The common targeted approach to control non-native plants may not always be effective, either ecologically or economically.

Cutting or mowing is used when herbicides are inappropriate, when benefits are limited due to growth stages of plant species or there are concerns about off-site effects. Cutting can control annual weeds, improve biocycling, increase tillering in some grasses and reduce seed production (Nuzzo 1991; Wilson & Clark 2001). It provides a selective advantage to defoliation-tolerant species and can eliminate some species by breaking their life cycle. Cutting increased growth and seed production of native species in grassland restoration (Maron & Jefferies 2001; Kimball et al. 2014); although it is not known if native species respond in the same way in all prairie types. Steaming uses a high pressure, hot water steam, finely sprayed close to the ground to kill vegetation and seeds (Merfield et al. 2009). Although effective in organic and cropping

agriculture, landscaping and forests (Norberg et al. 1997; Kolberg & Wiles 2002; Melander et al. 2005), it has not been studied in grassland restoration.

We investigated simple, cost-effective, non-native plant management methods for multiple species in ecological restoration of grasslands in Alberta, Canada. The objective was to determine whether cutting, glyphosate application and steaming, without targeted species, could reduce or eliminate non-native grasses and forbs with low impact to the non-target native plant community. We hypothesized glyphosate would most effectively reduce non-native plant cover since it can move into roots; cutting would be least damaging to native species as it is similar to grazing defoliation; and steam would provide short-term kill of above-ground vegetation.

Methods

Research sites

Waterton Lakes National Park is located in the Rocky Mountains of southwest Alberta, forming an International Peace Park and United Nations Educational, Scientific and Cultural Organization (UNESCO) World Heritage Site with neighbouring Glacier National Park in Montana, USA. The foothills grasslands growing season is cool and wet in May and June then hot and dry in August and September. Mean 1990–2000 annual precipitation is 807.6 mm, with most occurring during winter; mean minimum and maximum temperatures are -1.3 and 10.6 °C, respectively (Environment Canada 2012). The region is windy, with $70\text{--}90\text{ km}\cdot\text{h}^{-1}$ maximum daily gusts. Soils are Chernozemic with predominantly calcareous parent material, good drainage and dark coloured mineral surface horizons high in organic matter (Achuff et al. 2002).

Research sites were within a 3-km radius, with abundant non-native annual and perennial species representing typical grassland disturbances that could not be addressed with unique management techniques due to time and cost constraints. Site 1 was a landfill for domestic waste between 1952 and 1999. Soil was sandy loam texture with mean pH 7.5 and C:N ratio 15. Sites 2 and 3 were borrow pits for gravel quarries abandoned in the 1970s. Site 2 soil was sandy loam texture with mean pH 5.9 and C:N ratio 24. Site 3 soil was sand to sandy loam texture with mean pH 7.9 and C:N ratio 14. Soil in reference undisturbed fescue grassland had more total N, total P and organic C than adjacent disturbed areas, with lower pH at Sites 1 and 3 and higher at Site 2.

Experimental design and treatments

A randomized incomplete block design was used. In summer 2010, 0.3 ha at each site (blocks) was divided evenly

into 12 plots with uniform topography and vegetation. Non-native management treatments of cutting, glyphosate, steam and no treatment (control) were each randomly assigned to three replicate plots at each site. Steam was not used at Site 2 due to limited vehicle access. At each site, an undisturbed foothills fescue prairie reference plot adjacent to treatment plots was used to evaluate recovery of native plant communities following non-native plant management treatments.

Treatments were implemented in late July 2010. Hand-held grass trimmers were used to cut vegetation low to the ground; cuttings were not removed since non-native species were not in seed. A 3% solution (volume) of Roundup Transorb® (540 g l⁻¹ active ingredient glyphosate) was applied from backpack sprayers with pressure to maintain a fine spray at 15 to 20 l-plot⁻¹. Steam was applied by a commercial operator, dragging a 50-cm long metal bar across vegetation with steam released from narrow openings at 325 °C and 800 psi.

In 2011 treatments were applied in May when non-native species were most vulnerable. A 5% solution (volume) of Roundup® Concentrated Grass and Weed Control (143 g l⁻¹ active ingredient glyphosate) was applied using 2010 methods. Cutting was conducted in June using 2010 methods. Steam was applied at 225 °C and 350 psi by a commercial operator in June, with a hose approximately 50 cm above the vegetation (2010 operator ceased business). Plots were revegetated following these non-native plant management treatments as part of Parks Canada's standard approach to prevent further invasion by non-native species on active restoration sites, not for research purposes. An equal amount of seed (in 2011) and transplants (in 2012) were planted in all of the plots. Native species establishing from seed and transplants became part of the native vegetation in plots, which included the resident native vegetation and seed bank.

Soil assessment

In September 2011, three 500-g samples of soil were taken from 15 cm holes from each treatment and reference at each site, and stored at 4 °C until analysed by a commercial laboratory. Total C was determined by dry combustion (Nelson & Sommers 1996); inorganic C by CO₂ release (Loeppert & Suarez 1996); total organic C by calculation; organic matter by loss on ignition (Nelson & Sommers 1996); total N by Dumas method (Bremner 1996); total P by strong acid extraction and inductively coupled plasma mass spectrometry (US Environmental Protection Agency 1996); particle size by hydrometer (Kroetsch & Wang 2008); electrical conductivity and pH by saturated paste and meter (Miller & Curtin 2008); and Na adsorption ratio,

available Ca, Mg, Na and K by saturated paste and flame atomic adsorption spectroscopy (Miller & Curtin 2008).

Vegetation assessments

Vegetation was assessed in July 2010 to provide baseline data prior to treatment implementation. In subsequent years (2011, 2012 and 2013), vegetation was assessed in the last week of July or first week of August, when at peak production. Ten 0.1-m² quadrats were systematically located along two transects traversing each plot. In each quadrat, an ocular estimate of percentage canopy cover by species and percentage ground cover (bare ground, moss, vegetation, litter, rock) was made. Plant physiological stage of development was determined in 2011, with a single value assigned per quadrat to individual species. A score of 1 was assigned to immature plants (rosette stage or without reproductive parts), 2 to plants close to or flowering and 3 to plants with set seed.

Data analyses

Species were grouped into non-native forbs, non-native grasses, native forbs, native grasses and native shrubs; mean cover, physiological score and richness were calculated per plot. Native shrubs had very low density and cover, so were not included in analyses. Shannon's index of diversity was calculated for each plot and included native and non-native species. Cover of non-native species of concern in foothills fescue grasslands were compared among treatments.

Vegetation parameters were compared before and after treatments and year-to-year. Prior to non-native plant management, assessments were conducted in early July 2010. The first application of control methods was in late July 2010 and the second application late May–early June 2011. All plots were seeded in June 2011 and planted in June 2012. Post-treatment vegetation assessments were conducted in late July or early August 2011 (after two applications and seeding), 2012 (after two applications and planting) and 2013.

SAS version 9.2 (SAS Institute, Cary, NC, US) was used with alpha 0.05. All data were assessed for normality and homogeneity of variance using Shapiro-Wilk and Levene's tests, respectively.

To determine if site would influence outcomes, a two-way ANOVA was performed for each response variable within each year using proc glm with site and treatment as factors. Site × treatment effects were significant (data not shown), therefore data were analysed separately by site for treatment effects. Response variables (plant cover, physiological score, species richness, diversity) were compared among treatments with one-way ANOVA using proc

mixed with plot specified in the random statement. For significant ANOVAs, Tukey's *t*-tests were used to compare treatment means. Paired *t*-tests were conducted on cover and richness of plant groups and cover of individual non-native species of interest between pre- (2010) and final post-treatment year (2013).

Multivariate analyses were conducted to determine how site-specific soil conditions and plant community composition affected non-native plant control treatments with PC-ORD v 6 (MjM Software, Gleneden Beach, OR, US). Due to numerous zeros and heterogeneity of data, non-metric multidimensional scaling (NMDS) was used with the Sørensen distance measure, with 500 iterations in the final run, two axes and a random starting configuration. Dimensionality of the NMDS ordination was assessed using the stress test (Peck 2010). Correlations between key variables and ordination axes were tested with Spearman's rank correlation. Distinct plant community groups associated with treatments and sites were identified in the ordination and tested for significant differences using multiple response permutation procedure (MRPP) since sample units among groups were unequal. Indicator species analysis was used to confirm plant species characterizing treatments and sampling locations, based on constancy and distribution of abundance (Peck 2010).

Results

Plant abundance, diversity, flowering and seed production

In 2010 pre-treatment vegetation cover (Fig. 1), richness and diversity (Table 1) did not differ among plots within a site, except Site 2 where non-native forb richness was highest in the cutting treatment. Non-native grasses dominated Sites 1 and 2 and native forbs dominated Site 3 (Fig. 1). Total species richness and diversity were highest at Site 3 (Table 1). Non-native species abundance varied with site, with most species present at all sites pre-treatment. Dominant species were *Bromus inermis* (mean: 7.5% cover in plots) and *Elymus repens* (L.) Gould (quack grass; 9.2%) at Site 1, *B. inermis* (9.0%) and *Agropyron cristatum* (3.5%) at Site 2, and *Poa compressa* L. (Canada blue grass; 3.3%) and *A. cristatum* (2.8%) at Site 3.

Cutting and steam had little significant impact on vegetation diversity and species richness (Table 1). They did not reduce non-native grass and forb cover relative to controls within a year (Fig. 1), although over 3 yr, non-native grass cover was significantly lower with steam at Site 1 and with cutting at Site 3 (Fig. 2).

Glyphosate had significant impacts on vegetation in each year at each site. Non-native grass cover was reduced with glyphosate (Fig. 1). Although only statistically significant at Site 1 in 2012, this was biologically significant in all

years at all sites. Non-native grass cover with glyphosate was 75% less than the control at Site 2 in 2012, and 50% at Site 1 and 84% at Site 2 in 2013. Non-native grass species richness and diversity were significantly lower with glyphosate at Sites 1 and 2 in 2011 (Table 1) and at the end of the study relative to pre-treatment (Fig. 2). Native grass cover was <5% at all sites and did not significantly differ among treatments in any year. Total diversity was significantly lower with glyphosate across sites in 2011 but only at Site 3 in 2012.

Prior to and immediately after treatment, non-native forb cover was <5% at all sites and treatments, except significantly lower with glyphosate at Site 1 (Fig. 1). In 2012, non-native forb cover significantly increased with glyphosate at all sites, persisting in 2013 at two sites. Non-native forb richness was significantly higher with glyphosate at Sites 1 and 2 in 2012 and 2013 (Table 1). In 2011 and 2012 native forb cover and richness were significantly lower with glyphosate than controls at Site 3; but only cover at Site 2 in 2011 and Site 3 in 2013 (Fig. 1). The native forb species *Heterotheca villosa* (Pursh) Shinnars (hairy false golden aster) and *Achillea millefolium* L. (common yarrow) were eliminated by glyphosate. Native forbs were least negatively impacted at Site 1 where pre-treatment cover was very low. In 2013 the significantly higher native forb cover with glyphosate at Site 1 was due to increased *Verbena bracteata* Cav. ex Lag. & Rodr. (prostrate vervain), only found at this site. By the end of the study non-native forb cover was significantly higher with glyphosate than pre-treatment at Sites 1 and 2, and native forb cover had significantly decreased with glyphosate at Site 3 (Figs 1 and 2).

Flowering and seed set of non-native grasses was significantly reduced with glyphosate at Site 1, and that of non-native forbs was significantly reduced with cutting and glyphosate at Site 2 (Fig. 3). At Site 3, treatment had no effect on non-native plant flowering and seed production, although it significantly reduced that of native grasses and forbs.

Effect on non-native species

Individual non-native plant species had variable responses to treatments, with high variability within a site. Cutting and steam had no effect on any species. Within sites, *A. cristatum* (Site 1), *Poa pratensis* L. (Kentucky blue grass) and *P. compressa* (Site 2) were significantly reduced with glyphosate. *P. pratensis* and *E. repens* were eliminated from Site 3. Treatment effects on individual species were most significant in 2011, mainly between glyphosate and the control (Table 2). *P. compressa* at all sites, and *B. inermis* and *P. pratensis* at Site 2, were significantly reduced with glyphosate relative to the control and most other

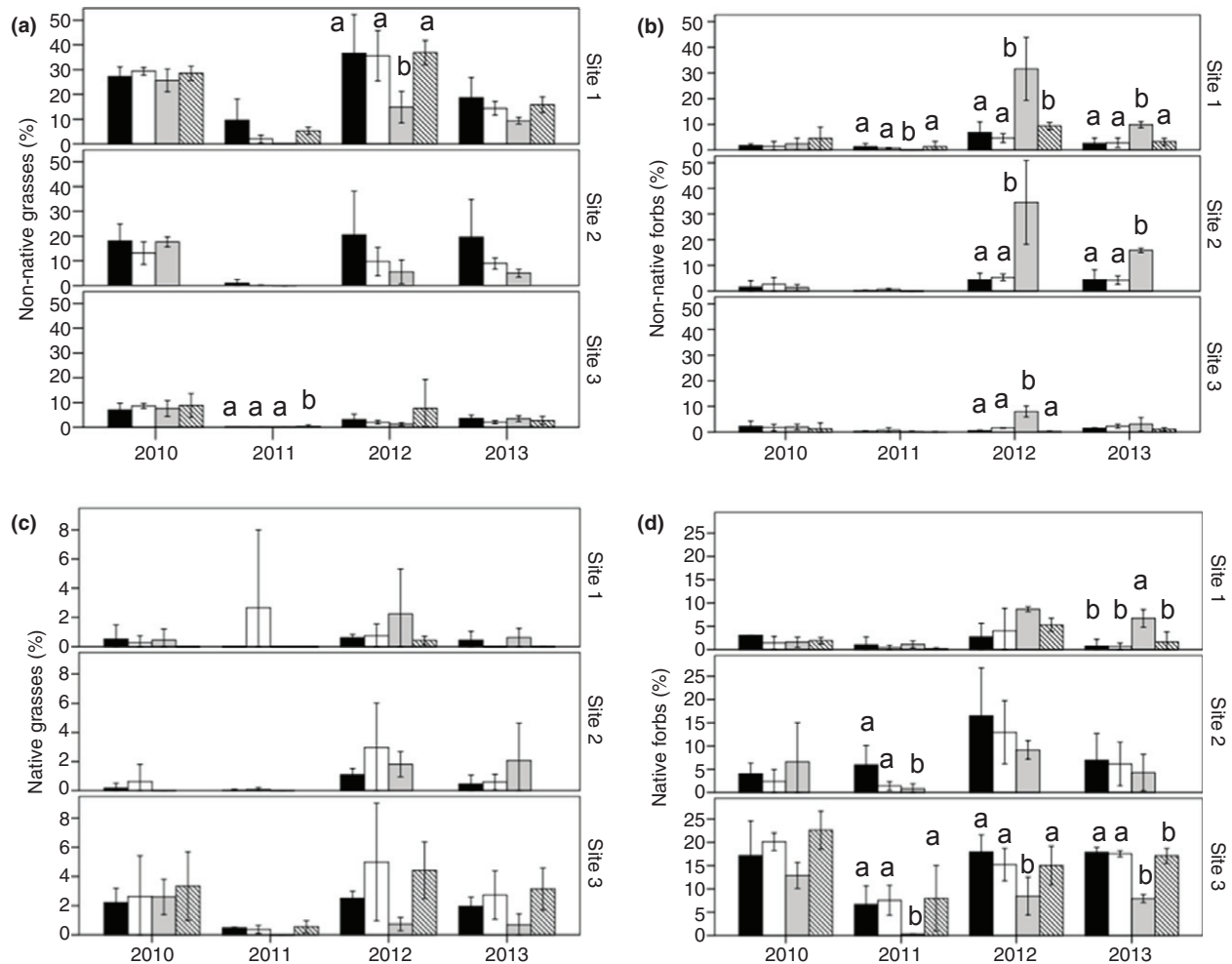


Fig. 1. Non-native (a,b) and native (c,d) mean vegetation cover at three restoration sites prior to (2010) and following (2011 to 2013) treatments to manage non-native plants. Treatments are control (black), cut (white), glyphosate (grey) and steam (hash). Bars represent ± 2 standard error. Columns within years that do not share a common letter are significantly different.

treatments. By 2013, there were no significant treatment effects on these species, suggesting non-native grasses had recovered.

At all sites, non-native forbs significantly increased with glyphosate in 2012, and remained significantly higher in 2013 at two sites (Fig. 1). Increased cover was mainly due to *Lepidium densiflorum* Scrad. (common pepper weed) and *Medicago lupulina* L. (black medic) at all sites and some species unique to sites, such as *Chrysanthemum leucanthemum* L. (ox-eye daisy) and *Verbascum thapsus* L. (common mullein) at Site 1 and *Cirsium arvense* (L.) Scop. (Canada thistle) at Site 2. *Centaurea maculosa* Lam. (spotted knapweed) (all three sites) and *C. leucanthemum*, species of high priority for control, were not reduced over time, and *B. tectorum* increased with glyphosate at Site 1 (data not shown).

Plant community development

Glyphosate shifted communities dominated by non-native perennial grasses to dominance by non-native forbs (Fig. 4a). Cutting and steam did not change plant community structure relative to the control. Site conditions explained the larger impact of glyphosate on native plant species at Sites 2 and 3. A gradient of increasing native plant species and decreasing non-native plant species was detected from Site 1 to Site 2 to Site 3, correlating with decreasing soil C:N ratio, clay and silt and increasing sand.

Comparing management treatments with target ecosystem data from undisturbed fescue grasslands showed distributions of individual native and non-native plant species were affected by soil properties contributing to the soil gradient (Fig. 4b). For example, *B. inermis* was significantly

Table 1. Mean plot species richness and diversity before and after non-native management at three restoration sites.

Shannon Diversity & Species Richness	Site 1				Site 2				Site 3			
	Control	Cut	Glyphosate	Steam	Control	Cut	Glyphosate	Steam	Control	Cut	Glyphosate	Steam
2010 (pre-treatment)												
Shannon Diversity	0.7	0.6	0.8	0.8	0.7	0.8	0.7	0.8	1.8	1.8	1.5	1.7
Native Forb	1 (0)	1 (1)	2 (1)	1 (1)	3 (1)	3 (0)	3 (1)	3 (1)	10 (0)	11 (1)	9 (0)	12 (2)
Native Graminoids	1 (1)	0 (0)	0 (0)	0 (0)	1 (1)	1 (1)	1 (1)	1 (1)	3 (0)	3 (0)	3 (1)	2 (0)
Non-native Forb	2 (1)	1 (1)	3 (1)	3 (1)	1 (0) ^a	3 (1) ^b	1 (0) ^a	1 (0) ^a	3 (1)	3 (1)	2 (1)	3 (1)
Non-native Grasses	4 (0)	4 (1)	5 (1)	4 (0)	4 (0)	4 (1)	4 (0)	4 (0)	2 (0)	4 (0)	3 (1)	3 (1)
2011												
Shannon Diversity	0.9 ^a	0.7 ^a	0.3 ^b	0.8 ^a	0.9 ^a	0.9 ^a	0.1 ^b	0.8 ^a	1.4 ^a	1.5 ^a	0.4 ^b	1.3 ^a
Native Forbs	1 (1)	2 (1)	3 (0)	2 (0)	6 (2)	6 (2)	3 (1)	2 (0)	15 (1) ^a	16 (1) ^a	7 (2) ^b	15 (2) ^a
Native Graminoids	3 (1)	1 (0)	0 (0)	1 (1)	1 (1)	3 (1)	1 (0)	1 (1)	3 (1)	2 (0)	1 (0)	3 (1)
Non-native Forbs	2 (1)	3 (0)	5 (1)	3 (1)	2 (0)	2 (1)	2 (1)	3 (1)	3 (1)	4 (1)	3 (1)	1 (0)
Non-native Grasses	6 (0) ^a	7 (0) ^a	3 (0) ^b	7 (1) ^a	5 (1) ^a	4 (1) ^a	1 (1) ^b	7 (1) ^a	4 (0)	3 (1)	2 (1)	4 (1)
2012												
Shannon Diversity	1.0	1.2	1.0	1.1	1.1	1.4	1.0	1.1	1.5 ^a	1.5 ^a	1.0 ^b	1.4 ^a
Native Forbs	3 (1)	2 (1)	4 (1)	2 (0)	11 (2)	11 (2)	9 (1)	2 (0)	20 (1) ^a	20 (1) ^a	11 (2) ^b	18 (3) ^a
Native Graminoids	3 (1)	1 (1)	3 (0)	2 (1)	3 (1)	5 (2)	4 (1)	2 (1)	3 (1)	4 (1)	4 (1)	4 (1)
Non-native Forbs	8 (1) ^a	7 (1) ^a	13 (2) ^b	7 (1) ^a	5 (0) ^a	7 (1) ^a	10 (1) ^b	7 (1) ^a	3 (1)	5 (1)	5 (1)	4 (1)
Non-native Grasses	8 (0)	7 (0)	8 (1)	7 (0)	4 (0)	4 (0)	4 (2)	7 (0)	3 (1)	4 (0)	2 (0)	4 (1)
2013												
Shannon Diversity	1.0	1.2	1.2	1.1	1.0	1.2	1.2	1.1	1.4	1.5	1.4	1.4
Native Forbs	1 (0)	1 (0)	2 (0)	1 (0)	6 (1)	6 (1)	4 (1)	1 (0)	13 (2)	15 (2)	10 (2)	14 (2)
Native Graminoids	2 (1)	0 (0)	1 (0)	1 (1)	2 (1)	2 (1)	2 (1)	1 (1)	3 (1)	4 (1)	3 (2)	2 (0)
Non-native Forbs	5 (1) ^{ab}	4 (1) ^a	7 (1) ^b	4 (1) ^a	4 (0) ^a	4 (0) ^a	6 (1) ^b	4 (1) ^a	3 (0)	4 (1)	4 (0)	3 (1)
Non-native Grasses	6 (0)	5 (1)	6 (0)	7 (0)	3 (0)	4 (0)	4 (0)	7 (0)	2 (0) ^a	4 (0) ^b	2 (0) ^a	2 (0) ^a

Numbers are mean ($\pm$ SE). Means followed by the same letter within a row and within a site are not significantly different.

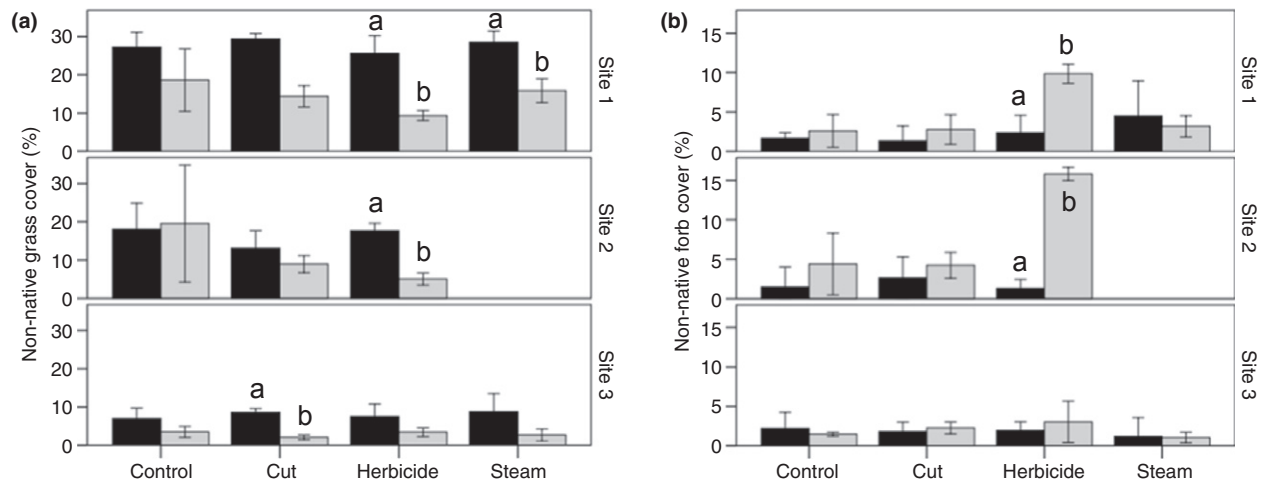


Fig. 2. Mean non-native (a) grass and (b) forb cover prior to (2010, black) and three years following (2013, grey) treatments to control non-native plants. Bars represent ± 2 standard error. Columns within treatments that do not share a common letter are significantly different.

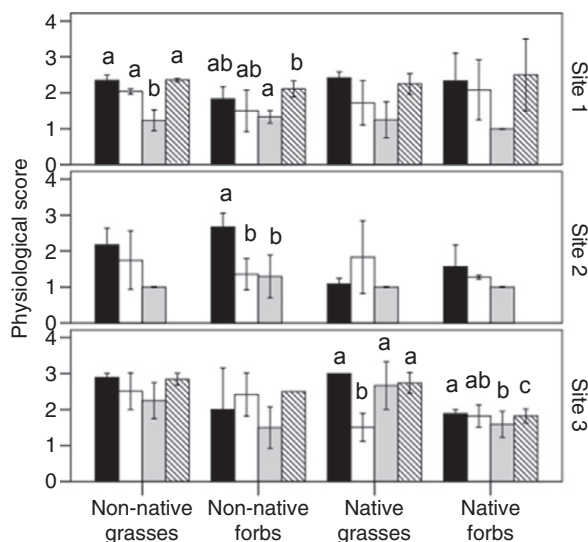


Fig. 3. Physiological score for plant groups in 2011 following non-native plant control at three restoration sites (1 = immature, 2 = close to flowering or flowering and 3 = seed set). Treatments are control (black), cut (white), glyphosate (grey) and steam (hash). Bars represent ± 2 standard error. Columns within plant groups that do not share a common letter are significantly different.

positively correlated with increasing soil clay content. Dominant non-native perennial grasses and *C. leucanthemum* were identified as strongly influencing Sites 1 and 2 plant community composition. As expected, undisturbed and disturbed communities were distinctly different; the former dominated by *Festuca campestris* Rydb. (rough fescue), *F. idahoensis* Elmer (Idaho fescue), *Danthonia parryi* Scribn. (Parry's oat grass) and *Selaginella densa* Rydb. (little

club moss), which were absent or with very low abundance in disturbed sites.

Discussion

Glyphosate applied without targeting specific species was an effective and easily implemented management technique to reduce non-native perennial grasses in the short term, and may be used to control species posing a significant threat to conservation of native biodiversity. The reduction of non-native grasses, but not non-native forbs, poses a continuing challenge to grassland restoration with multiple non-native species. Two early spring herbicide applications when grasses were establishing and putting resources into shoot production may reduce non-native grasses more with less harm to native forbs than in July when grasses, and forbs, are at peak production. Although non-native grass cover and flowering were reduced following treatment, other research has shown the effect may be temporary and repeated control required over the longer term (Rice et al. 1997; Holl et al. 2014). Non-native forbs are most effectively controlled by herbicide when actively allocating resources to both above- and below-ground tissues and before flowering and seed set occurs. Spraying the dominant grasses in spring with follow-up spraying to manage non-native forbs that establish and replace these grasses later in the growing season could be an effective management technique, although caution must be taken to protect native forbs.

Increased non-native forbs with glyphosate was likely due to a larger reduction in non-native grasses, resulting in release from competition for other plants; and/or to different species response to glyphosate (DiTomaso 2000;

Table 2. Effect of glyphosate application on cover of non-native species of concern in fescue grassland.

	Non-Native Grasses					Non-Native Forbs	
	<i>Agropyron cristatum</i>	<i>Bromus inermis</i>	<i>Elymus repens</i>	<i>Poa compressa</i>	<i>Poa pratensis</i>	<i>Lepidium densiflorum</i>	<i>Medicago lupulina</i>
Site 1							
2010	Control	3.3 (1.8)	5.3 (2.0)	4.3 (3.0)	7.3 (3.7) A	7.7 (1.7)	0.0 (0.0)
	Glyphosate	3.0 (0.0) A	7.0 (1.7)	8.3 (1.2)	5.7 (0.9) A	1.0 (0.0)	0.0 (0.0)
2011	Control	3.8 (2.6)	9.7 (3.4)	1.3 (0.6)	3.0 (0.5) a	0.9 (0.4)	0.0 (0.0)
	Glyphosate	0.0 (0.0)	0.3 (0.3)	1.4 (0.2)	0.1 (0.1) b	0.0 (0.0)	0.0 (0.0)
2012	Control	3.8 (2.6)	9.5 (2.2)	14.9 (7)	2.8 (0.5) a	1.9 (0.6)	0.0 (0.0)
	Glyphosate	0.5 (0.4)	1.9 (1.2)	8.5 (1.1)	0.6 (0.2) b	0.2 (0.1)	0.1 (0.04)
2013	Control	2.1 (1.6)	7.8 (3)	2.2 (0.9)	0.6 (0.3) B	3.5 (1.8)	0.0 (0.0)
	Glyphosate	1.1 (0.4) B	1.1 (0.6)	2.9 (0.8)	1.0 (0.3) B	0.4 (0.4)	0.2 (0.1)
Site 2							
2010	Control	4.3 (2.2)	8.0 (2.0)	0.7 (0.7)	1.3 (0.9)	2.0 (1.0)	0.0 (0.0)
	Glyphosate	9.0 (1.0)	5.3 (2.3)	1.3 (0.9)	7.3 (0.9) A	7.0 (1.7)	0.0 (0.0)
2011	Control	1.0 (0.6)	10.2 (3.4) a	0.0 (0.0)	1.9 (0.6) a	1.9 (0.4) a	0.0 (0.0)
	Glyphosate	0.0 (0.0)	0.3 (0.1) b	0.0 (0.0)	0.0 (0.0) b	0.0 (0.0) b	0.0 (0.0)
2012	Control	1.8 (0.4)	10.3 (3.5)	0.0 (0.0)	0.5 (0.2)	5.9 (3.7)	0.1 (0.03) a
	Glyphosate	1.2 (1.2)	2.9 (0.5)	0.0 (0.0)	0.2 (0.1)	0.0 (0.0)	11.3 (2.5) b
2013	Control	1.4 (0.7)	14.8 (6.4)	0.0 (0.0)	2.8 (2)	0.5 (0.5)	10.2 (4.5)
	Glyphosate	1.5 (0.7)	2.0 (0.9)	0.1 (0.1)	1.2 (0.5) B	0.0 (0.0)	2.3 (0.9)
Site 3							
2010	Control	1.3 (0.7)	1.3 (0.7)	0.0 (0.0)	1.7 (0.9)	1.3 (0.3)	0.0 (0.0)
	Glyphosate	1.3 (1.3)	1.7 (1.2)	1.7 (0.9)	1.3 (0.7)	2.7 (2.7)	0.0 (0.0)
2011	Control	0.6 (0.4)	0.2 (0.1)	0.0 (0.0)	2.4 (0.7) a	0.0 (0.0)	0.0 (0.0)
	Glyphosate	0.0 (0.0)	0.1 (0.1)	0.0 (0.0)	0.1 (0.1) b	0.1 (0.1)	0.0 (0.0)
2012	Control	2.0 (1.3)	0.1 (0.1)	0.0 (0.0)	1.0 (0.1)	0.0 (0.0)	0.1 (0.1) a
	Glyphosate	0.9 (0.2)	0.1 (0.1)	0.0 (0.0)	0.4 (0.2)	0.0 (0.0)	3.2 (1.1) b
2013	Control	1.9 (0.1)	0.2 (0.2)	0.0 (0.0)	1.4 (0.6)	0.0 (0.0)	0.0 (0.0)
	Glyphosate	1.8 (0.8)	0.0 (0.0)	0.0 (0.0)	1.6 (0.3)	0.0 (0.0)	0.6 (0.1)
Numbers are mean (±SE). Within a column and year, means with different lowercase letters are significantly different. Within sites and treatments, means with different uppercase letters are significantly different between 2010 (pre-treatments) and 2013.							

Numbers are mean ($\pm$ SE). Within a column and year, means with different lowercase letters are significantly different between 2010 (pre-treatments) and 2013.

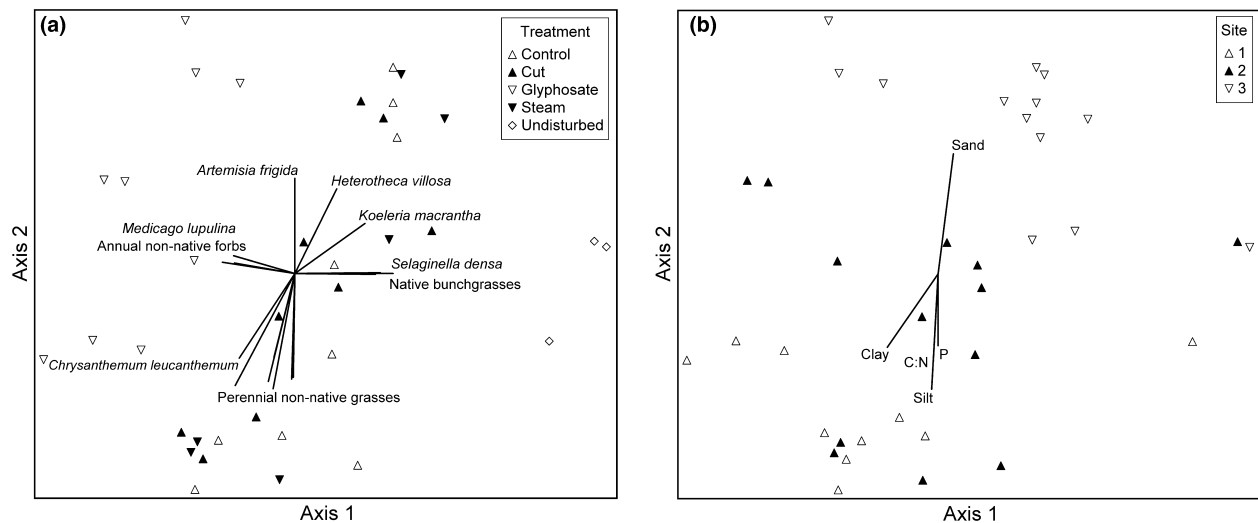


Fig. 4. Non-metric multidimensional ordination of plant communities from non-native species management treatments (a) and sites (b). Axis 1 accounted for 34.7% of the variation in data and Axis 2 48%. A final stress value of 14.42 was obtained for the ordination. Significant correlations with plant species and soil properties are labeled in the biplots (a) and (b), respectively. In (a), perennial non-native grasses are represented by the set of overlapping correlation axes to the right of *Chrysanthemum leucanthemum* and from left to right are: *Poa pratensis*, *Elymus repens*, *Phleum pratense*, *Bromus inermis* and *Agropyron cristatum*.

Murphy & Grant 2005; Simmons et al. 2007; Kuebbing & Nuñez 2015). Similar compensatory increases in non-native grasses occurred with clopyralid, a forb-specific herbicide to control *Centaurea maculosa* (Martin et al. 2014). Non-native forbs that increased in this study were annual weeds such as *Lepidium densiflorum* and *Medicago lupulina* and perennial invasive species such as *Cirsium arvense* and *Centaurea maculosa*. Although some research has shown annual non-native forbs can dominate during early succession, then gradually decrease without management (Gerling et al. 1996), in the longer term, establishment and spread of perennial invasive forbs may be detrimental to grassland restoration. At the site in early stages of invasion (Site 3), non-native forb abundance was declining in the final year but not at the other two more heavily invaded sites. Thus longevity of treatment effects for grasses and forbs should continue to be evaluated, as results to date are not conclusive. The effects of herbicide are often short-lived, so repeated herbicide application has been recommended for every 2–3 yr to prevent reinvasion (Holl et al. 2014). We only recommend this for heavily invaded sites, and suggest spot spraying may be a less detrimental alternative for native species.

Flory & Clay (2009) found multiple techniques controlled invasive species, although native plant community response varied with method. In contrast to our findings and others (e.g. Rinella et al. 2001; Orloff et al. 2015), native diversity and abundance increased following invasive species removal. In our study, native forbs did not establish on heavily invaded sites and declined as non-

native forbs increased at a less invaded site. Rinella et al. (2001) found key native grassland forbs were rare 16 yr after picloram control for non-native plants, and others found temporary declines in native species richness and diversity (Kaiser-Bunbury et al. 2015). When certain species are eliminated they may leave niches that can be filled by other non-native species (Pearson et al. 2016). Following broadcast methods to remove fennel in grasslands, the native community did not recover and non-native grasses readily established due to a lack of native seed bank and grass litter (Ogden & Rejmanek 2005). Both are factors in our study as the native seed bank in heavily invaded sites may have become depleted over past decades, and litter production in glyphosate and cutting treatments may inhibit native grass and forb establishment.

Two annual cuttings did not eliminate large and diverse populations of non-native plant species, similar to other short-term studies (Simmons et al. 2007). Short-term cutting is likely more effective with one non-native species to be controlled (Rinella et al. 2001), when repeated during the growing season, for longer periods of time and when tiller apices are removed (Willson & Stubbendieck 1996; Wilson & Clark 2001).

Our study was the first to use steam in grassland for non-native plant control. Steam only reduced non-native grasses at one site – the most heavily invaded. Inconsistent results may occur from heat loss due to absence of a steam insulation component (Labowsky 1990; Norberg et al. 1997), insufficient applications and too short a time period (<2 yr) for steam to significantly affect vegetation

(Kristoffersen et al. 2007). Additionally, perennial non-native grasses dominating our sites are difficult to kill with steam (Wei et al. 2010). Innovative strategies like steam are worth investigating further to address concerns regarding herbicide use in some environments, although are not feasible for sites with poor vehicle access.

Site variability influenced plant community response, with most restoration sites presenting different conditions and disturbance histories that led to unique outcomes following management. Wide site-to-site variability is increasingly recognized and has led to emergence of multi-site investigations, termed coordinated distributed experiments (Fraser et al. 2012; Borer et al. 2014). Age of disturbance was similar among sites; however, factors such as soil properties and adjacent disturbance and native and non-native plant abundances may direct level of non-native plant invasion prior to and post treatments (Orloff et al. 2015). Year-to-year variation in native and non-native species abundances in grasslands has been correlated with precipitation (Ogden & Rejmanek 2005), which may partially explain why responses varied among years.

Native grasslands are rapidly declining on the landscape and restoration is imperative. Our study showed that controlling multiple non-native species occurring with native species, rather than management of a single undesirable plant species, is not a simple task as site-specific conditions may influence control methods for non-native species. Having three sites with different histories, abiotic conditions and different initial plant community composition showed the importance of site specificity. Although no single treatment effectively re-established native grassland communities over 3 yr, glyphosate was effective for controlling non-native grasses, but induced a shift to non-native forb dominance. When native forbs were abundant prior to management, glyphosate reduced them. Herbicide should be used with caution at less invaded sites with higher native species diversity. With low frequency applications over 2 yr, cutting was not a useful management strategy for reducing non-native grasses and forbs, although steam may have potential since it reduced non-native grasses at one site. Further work should investigate a follow-up herbicide treatment to manage non-native forbs that replace non-native grasses in invaded sites.

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